Faculté des Sciences Aïn Chock — Master Big Data et Cloud Computing (BD2C)
Module : Virtualisation et Cloud Computing — 2026/2027

Compte rendu — Exercices 1 à 12

Captures, résultats et réponses aux questions

Ce compte rendu regroupe, exercice par exercice, les commandes exécutées et leurs résultats (captures d'écran légendées).

Sommaire : Exercice 1 à Exercice 12 (une section par exercice).

Exercice 1

Vérification des outils, du réseau libvirt et des machines existantes.

```
zahra@ubuntu:~/tp1-kvm/formats$ command -v genisoimage; command -v virt-install; command -v qemu-img
sudo virsh net-list --all
mkdir -p ~/bd2c/tp2/{nocloud,templates,profiles,inventory,generated}
ls ~/bd2c/tp2
/usr/bin/genisoimage
/usr/bin/virt-install
/usr/bin/qemu-img
 Name      State    Autostart   Persistent
--------------------------------------------
 default   active   yes         yes

generated  inventory  nocloud  profiles  templates
zahra@ubuntu:~/tp1-kvm/formats$
```

Figure 1 — Vérification des outils (genisoimage, virt-install, qemu-img), du réseau libvirt « default » et création de l'arborescence ~/bd2c/tp2.

```
zahra@ubuntu:~/tp1-kvm/formats$ sudo virsh domifaddr ubuntu-app01 --source lease
 Name       MAC address          Protocol     Address
-------------------------------------------------------------------------------
 vnet0      52:54:00:5d:0b:e2    ipv4         192.168.122.236/24

zahra@ubuntu:~/tp1-kvm/formats$ sudo virsh net-dhcp-leases default
 Expiry Time           MAC address         Protocol   IP address           Hostname   Client ID or DUID
-------------------------------------------------------------------------------------------------------------------------------------------------
 2026-11-05 18:22:05   52:54:00:5d:0b:e2   ipv4       192.168.122.236/24   app01      ff:56:50:4d:98:00:02:00:00:ab:11:a8:2c:22:aa:77:1a:e7:5e
 2026-11-05 18:19:43   52:54:00:74:ec:03   ipv4       192.168.122.251/24   app02      ff:56:50:4d:98:00:02:00:00:ab:11:14:1b:3d:66:e6:25:58:29
```

Figure 2 — Adresses IP et baux DHCP des instances ubuntu-app01 et ubuntu-app02 (réseau default).

Exercice 2

Examen de cloud-init dans une instance et premier fichier user-data.

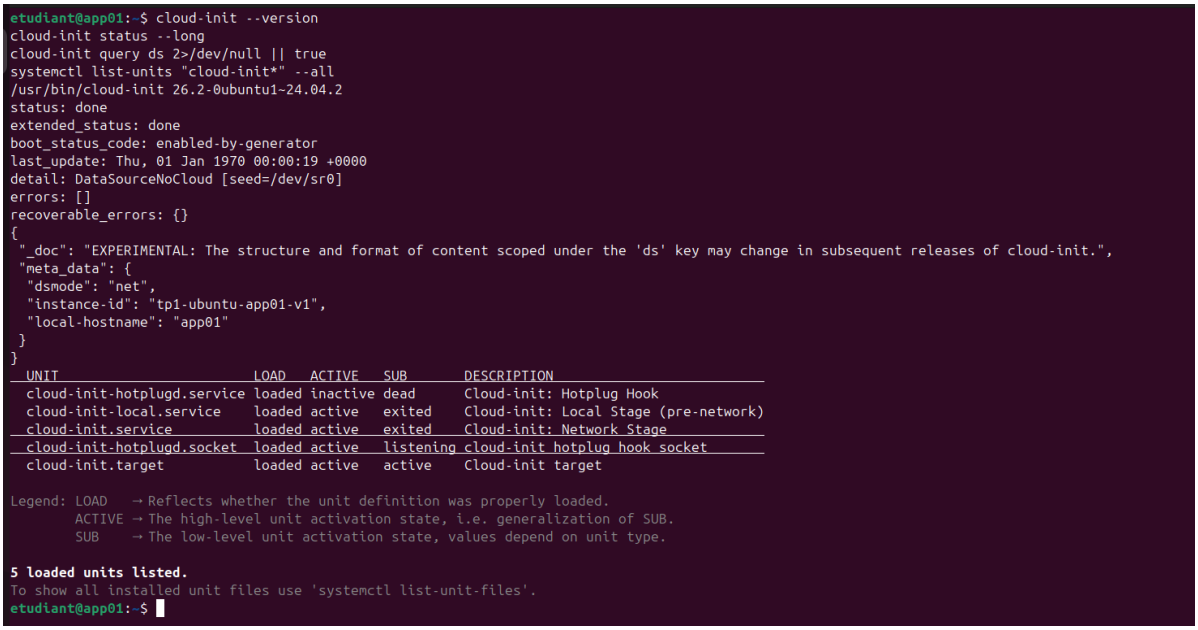


Figure 3 — Version et état de cloud-init dans ubuntu-app01 (datasource NoCloud, status: done, unités cloud-init actives).



Figure 4 — Création du fichier user-data (hostname bd2c-vm01, utilisateur etudiant) et validation de la syntaxe YAML.

Exercice 3

Création des fichiers user-data et meta-data, puis de l'image seed (volume cidata).

```
zahra@ubuntu:~/bd2c/tp2/nocloud$ cat > user-data <<'EOF'
#cloud-config
hostname: bd2c-vm01
manage_etc_hosts: true
users:
  - default
  - name: etudiant
    groups: [sudo]
    shell: /bin/bash
    sudo: "ALL=(ALL) NOPASSWD:ALL"
    lock_passwd: true
EOF
zahra@ubuntu:~/bd2c/tp2/nocloud$ cat > meta-data <<'EOF'
instance-id: iid-bd2c-vm01
local-hostname: bd2c-vm01
EOF
```

Figure 5 — Création de user-data et de meta-data (instance-id : iid-bd2c-vm01).

```
zahra@ubuntu:~/bd2c/tp2/nocloud$ ls -l
cat user-data
cat meta-data
python3 -c "import yaml; yaml.safe_load(open('user-data')); print('YAML OK')"
total 8
-rw-rw-r-- 1 zahra zahra  53 nov.   5 18:03 meta-data
-rw-rw-r-- 1 zahra zahra 192 nov.   5 18:02 user-data
#cloud-config
hostname: bd2c-vm01
manage_etc_hosts: true
users:
  - default
  - name: etudiant
    groups: [sudo]
    shell: /bin/bash
    sudo: "ALL=(ALL) NOPASSWD:ALL"
    lock_passwd: true
instance-id: iid-bd2c-vm01
local-hostname: bd2c-vm01
YAML OK
zahra@ubuntu:~/bd2c/tp2/nocloud$
```

Figure 6 — Contenu des fichiers et validation YAML (« YAML OK »).

```
zahra@ubuntu:~/bd2c/tp2/nocloud$ genisoimage -output ../seed-vm01.iso -volid cidata -joliet -rock user-data meta-data
ls -lh ../seed-vm01.iso
I: -input-charset not specified, using utf-8 (detected in locale settings)
Total translation table size: 0
Total rockridge attributes bytes: 331
Total directory bytes: 0
Path table size(bytes): 10
Max brk space used 0
183 extents written (0 MB)
-rw-rw-r-- 1 zahra zahra 366K nov.   5 18:04 ../seed-vm01.iso
zahra@ubuntu:~/bd2c/tp2/nocloud$ isoinfo -d -i ../seed-vm01.iso | grep -i "volume id"
isoinfo -R -f -i ../seed-vm01.iso
Volume id: cidata
/meta-data
/user-data
zahra@ubuntu:~/bd2c/tp2/nocloud$
```

Figure 7 — Génération de seed-vm01.iso (volume « cidata ») et vérification avec isoinfo.

Exercice 4

Déploiement d'une VM à partir du template avec le seed.

```
zahra@ubuntu:~/bd2c/tp2/nocloud$ sudo ls -lh /var/lib/libvirt/images/
sudo virsh net-list --all
[sudo] Mot de passe de zahra :
total 24G
-rw-r--r-- 1 libvirt-qemu kvm  3,3G nov.   5 19:11 ubuntu-app01.qcow2
-rw-r--r-- 1 libvirt-qemu kvm  3,2G nov.   5 19:10 ubuntu-app02.qcow2
-rw-r--r-- 1 root         root 3,5G oct.  27 20:41 ubuntu-os-capture-v1.qcow2
-rw-r--r-- 1 root         root 3,8G nov.   3 19:28 ubuntu-restore01.qcow2
-rw-r--r-- 1 root         root 3,2G nov.   4 23:10 ubuntu-template-v1.qcow2
-rw------- 1 root         root  16G nov.   4 23:02 ubuntu-vm01.qcow2
-rw------- 1 root         root 4,3G oct.  27 14:36 ubuntu-vm02.qcow2
 Name      State    Autostart   Persistent
----------------------------------------------
 default   active   yes         yes
```

Figure 8 — Contenu de /var/lib/libvirt/images/ : template, disques des VM existantes.

```
zahra@ubuntu:~/bd2c/tp2/nocloud$ sudo virsh list --all
sudo qemu-img info /var/lib/libvirt/images/ubuntu-template-v1.qcow2
 Id   Name              State
--------------------------------
 1    ubuntu-app01      running
 2    ubuntu-app02      running
 -    ubuntu-restore01  shut off
 -    ubuntu-vm01       shut off
 -    ubuntu-vm02       shut off

image: /var/lib/libvirt/images/ubuntu-template-v1.qcow2
file format: qcow2
virtual size: 15 GiB (16106127360 bytes)
disk size: 3.14 GiB
cluster_size: 65536
Format specific information:
    compat: 1.1
    compression type: zlib
    lazy refcounts: false
    refcount bits: 16
    corrupt: false
    extended l2: false
Child node '/file':
    filename: /var/lib/libvirt/images/ubuntu-template-v1.qcow2
    protocol type: file
    file length: 3.14 GiB (3374776320 bytes)
    disk size: 3.14 GiB
zahra@ubuntu:~/bd2c/tp2/nocloud$ sudo apt install -y libguestfs-tools
sudo virt-ls -a /var/lib/libvirt/images/ubuntu-template-v1.qcow2 /usr/bin | grep cloud-init
Lecture des listes de paquets... Fait
Construction de l'arbre des dépendances... Fait
Lecture des informations d'état... Fait
libguestfs-tools est déjà la version la plus récente (1:1.52.0-5ubuntu3).
0 mis à jour, 0 nouvellement installés, 0 à enlever et 343 non mis à jour.
cloud-init
cloud-init-per
zahra@ubuntu:~/bd2c/tp2/nocloud$
```

Figure 9 — État des VM, informations sur ubuntu-template-v1.qcow2 et présence de cloud-init dans le template (virt-ls).

```
zahra@ubuntu:~/bd2c/tp2/nocloud$ sudo cp ~/bd2c/tp2/seed-vm01.iso /var/lib/libvirt/images/
sudo cp /var/lib/libvirt/images/ubuntu-template-v1.qcow2 /var/lib/libvirt/images/bd2c-vm01.qcow2
sudo ls -lh /var/lib/libvirt/images/ | grep -E "bd2c|seed"
-rw-r--r-- 1 root         root 3,2G nov.   5 19:19 bd2c-vm01.qcow2
-rw-r--r-- 1 root         root 366K nov.   5 19:18 seed-vm01.iso
zahra@ubuntu:~/bd2c/tp2/nocloud$ free -h
               total       utilisé      libre     partagé tamp/cache   disponible
Mem:           7,7Gi       3,4Gi       155Mi        40Mi       4,5Gi       4,3Gi
Échange:       3,8Gi       1,6Mi       3,8Gi
zahra@ubuntu:~/bd2c/tp2/nocloud$ sudo virt-install --connect qemu:///system \
  --name bd2c-vm01 --memory 2048 --vcpus 2 \
  --disk path=/var/lib/libvirt/images/bd2c-vm01.qcow2,format=qcow2,bus=virtio \
  --disk path=/var/lib/libvirt/images/seed-vm01.iso,device=cdrom \
  --network network=default,model=virtio \
  --import --os-variant generic --graphics none --noautoconsole
WARNING  Using --osinfo generic, VM performance may suffer. Specify an accurate OS for optimal results.

Début d'installation…
Création du domaine…
Création de domaine terminée.
zahra@ubuntu:~/bd2c/tp2/nocloud$ v
v : commande introuvable
zahra@ubuntu:~/bd2c/tp2/nocloud$ sudo virsh list --all
sudo virsh domifaddr bd2c-vm01 --source lease
 Id   Name              State
---------------------------------
 1    ubuntu-app01      running
 2    ubuntu-app02      running
 3    bd2c-vm01         running
 -    ubuntu-restore01  shut off
 -    ubuntu-vm01       shut off
 -    ubuntu-vm02       shut off

 Name       MAC address          Protocol     Address
-------------------------------------------------------------------------------
 vnet2      52:54:00:7e:72:1a    ipv4         192.168.122.16/24
```

Figure 10 — Copie du template et du seed, création de bd2c-vm01 avec virt-install, puis adresse IP obtenue.

Exercice 5

Accès SSH par clé injectée avec cloud-init.

```
zahra@ubuntu:~/bd2c/tp2/nocloud$ ssh-keygen -t ed25519 -f ~/.ssh/bd2c_tp2 -C bd2c-tp2
cat ~/.ssh/bd2c_tp2.pub
Generating public/private ed25519 key pair.
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/zahra/.ssh/bd2c_tp2
Your public key has been saved in /home/zahra/.ssh/bd2c_tp2.pub
The key fingerprint is:
SHA256:RXFTQ5iU6pWKSZc+wdsXl7l8iB004PMgTwevnlcIDow bd2c-tp2
The key's randomart image is:
+--[ED25519 256]--+
|          oo*B+  |
|         + o++o. |
|        E.=o*oo.o|
|        ..**oBo+.|
|       .S* *+o+=o|
|        o *.o.++.|
|           .o.. .|
|             .   |
|                 |
+----[SHA256]-----+
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIBunbClBfIv91HO2+WiYpAMS3U/2Z96gHA0I5H5lM2am bd2c-tp2
zahra@ubuntu:~/bd2c/tp2/nocloud$
```

Figure 11 — Génération de la paire de clés SSH dédiée bd2c_tp2 (ed25519) et affichage de la clé publique.

```
zahra@ubuntu:~/bd2c/tp2/nocloud$ cd ~/bd2c/tp2/nocloud
PUBKEY=$(cat ~/.ssh/bd2c_tp2.pub)
cat > user-data <<EOF
#cloud-config
hostname: bd2c-vm01
manage_etc_hosts: true
users:
  - default
  - name: etudiant
    shell: /bin/bash
    groups: [sudo]
    sudo: "ALL=(ALL) NOPASSWD:ALL"
    lock_passwd: true
    ssh_authorized_keys:
      - ${PUBKEY}
ssh_pwauth: false
EOF
cat user-data
python3 -c "import yaml; yaml.safe_load(open('user-data')); print('YAML OK')"
#cloud-config
hostname: bd2c-vm01
manage_etc_hosts: true
users:
  - default
  - name: etudiant
    shell: /bin/bash
    groups: [sudo]
    sudo: "ALL=(ALL) NOPASSWD:ALL"
    lock_passwd: true
    ssh_authorized_keys:
      - ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIBunbClBfIv91HO2+WiYpAMS3U/2Z96gHA0I5H5lM2am bd2c-tp2
ssh_pwauth: false
YAML OK
zahra@ubuntu:~/bd2c/tp2/nocloud$
```

Figure 12 — user-data enrichi avec la clé publique (ssh_authorized_keys, ssh_pwauth: false) et validation YAML.

```
zahra@ubuntu:~/bd2c/tp2/nocloud$ cd ~/bd2c/tp2/nocloud
sed -i 's/iid-bd2c-vm01$/iid-bd2c-vm01-v2/' meta-data
genisoimage -output ../seed-vm01-v2.iso -volid cidata -joliet -rock user-data meta-data
sudo cp ../seed-vm01-v2.iso /var/lib/libvirt/images/

sudo virsh destroy bd2c-vm01
sudo virsh undefine bd2c-vm01
sudo cp /var/lib/libvirt/images/ubuntu-template-v1.qcow2 /var/lib/libvirt/images/bd2c-vm01.qcow2

sudo virt-install --connect qemu:///system \
  --name bd2c-vm01 --memory 2048 --vcpus 2 \
  --disk path=/var/lib/libvirt/images/bd2c-vm01.qcow2,format=qcow2,bus=virtio \
  --disk path=/var/lib/libvirt/images/seed-vm01-v2.iso,device=cdrom \
  --network network=default,model=virtio \
  --import --os-variant generic --graphics none --noautoconsole

sudo virsh domifaddr bd2c-vm01 --source lease
ssh -i ~/.ssh/bd2c_tp2 etudiant@IP
I: -input-charset not specified, using utf-8 (detected in locale settings)
Total translation table size: 0
Total rockridge attributes bytes: 331
Total directory bytes: 0
Path table size(bytes): 10
Max brk space used 0
183 extents written (0 MB)
Domain 'bd2c-vm01' destroyed

Domain 'bd2c-vm01' has been undefined

WARNING  Using --osinfo generic, VM performance may suffer. Specify an accurate OS for optimal results.

Début d'installation…
Création du domaine…
Création de domaine terminée.
 Name       MAC address          Protocol     Address
-------------------------------------------------------------------------------

ssh: Could not resolve hostname ip: Name or service not known
```

Figure 13 — Nouveau seed (instance-id iid-bd2c-vm01-v2) et recréation de bd2c-vm01. La connexion SSH échoue : l'adresse IP n'avait pas été substituée dans la commande (« Could not resolve hostname ip »).

```
zahra@ubuntu:~/bd2c/tp2/nocloud$ sudo virsh domifaddr bd2c-vm01 --source lease
 Name       MAC address          Protocol     Address
-------------------------------------------------------------------------------
 vnet3      52:54:00:c9:f2:ef    ipv4         192.168.122.20/24
```

Figure 14 — Relevé de l'adresse IP de la VM (domifaddr).

```
zahra@ubuntu:~/bd2c/tp2/nocloud$ ssh -i ~/.ssh/bd2c_tp2 etudiant@192.168.122.20
The authenticity of host '192.168.122.20 (192.168.122.20)' can't be established.
ED25519 key fingerprint is SHA256:d0AKsvH0uRpa+6O3gTk1AoQCB+eQuWu7BC0enixSyNs.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.122.20' (ED25519) to the list of known hosts.
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-147-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Thu Nov  5 06:38:23 PM UTC 2026

  System load:  0.09               Processes:             114
  Usage of /:   32.2% of 14.66GB   Users logged in:       0
  Memory usage: 8%                 IPv4 address for ens2: 192.168.122.20
  Swap usage:   0%

 * Canonical Workshop gives developers fast, composable, reproducible, and
   secure developer environments that are perfect for agentic workflows.

   https://ubuntu.com/workshop

Expanded Security Maintenance for Applications is not enabled.

62 updates can be applied immediately.
1 of these updates is a standard security update.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

New release '26.04.1 LTS' available.
Run 'do-release-upgrade' to upgrade to it.
```

Figure 15 — Connexion SSH par clé à etudiant@192.168.122.20.

```
etudiant@bd2c-vm01:~$ hostnamectl
cloud-init status --wait
 Static hostname: bd2c-vm01
       Icon name: computer-vm
         Chassis: vm
      Machine ID: 759360de8c844ea98e43da4dfb75f976
         Boot ID: 27a578f851584d239bf9c7556d9a8137
  Virtualization: kvm
Operating System: Ubuntu 24.04.4 LTS
          Kernel: Linux 6.8.0-147-generic
    Architecture: x86-64
 Hardware Vendor: QEMU
  Hardware Model: Ubuntu 24.04 PC v2 _i440FX + PIIX, arch_caps fix, 1996_
Firmware Version: 1.16.3-debian-1.16.3-2
   Firmware Date: Tue 2014-04-01
    Firmware Age: 12y 7month 5d
status: done
etudiant@bd2c-vm01:~$
```

Figure 16 — hostnamectl dans la VM : nom d'hôte bd2c-vm01 ; cloud-init status: done.

Exercice 6

Configuration réseau : DHCP puis adresse IP statique avec network-config.

```
etudiant@bd2c-vm01:~$ ip addr
ip route
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: ens2: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 10
    link/ether 52:54:00:c9:f2:ef brd ff:ff:ff:ff:ff:ff
    altname enp0s2
    inet 192.168.122.20/24 metric 100 brd 192.168.122.255 scope global dynamic ens2
       valid_lft 3240sec preferred_lft 3240sec
    inet6 fe80::5054:ff:fec9:f2ef/64 scope link
       valid_lft forever preferred_lft forever
default via 192.168.122.1 dev ens2 proto dhcp src 192.168.122.20 metric 100
192.168.122.0/24 dev ens2 proto kernel scope link src 192.168.122.20 metric 100
192.168.122.1 dev ens2 proto dhcp scope link src 192.168.122.20 metric 100
```

Figure 17 — Configuration réseau obtenue par DHCP dans la VM (ip addr et ip route : 192.168.122.20).

```
zahra@ubuntu:~/bd2c/tp2/nocloud$ cd ~/bd2c/tp2/nocloud
cat > network-config <<'EOF'
version: 2
ethernets:
  ens2:
    dhcp4: true
EOF
sed -i 's/^instance-id:.*/instance-id: iid-bd2c-vm01-v3/' meta-data
genisoimage -output ../seed-reseau.iso -volid cidata -joliet -rock user-data meta-data network-config
sudo cp ../seed-reseau.iso /var/lib/libvirt/images/
I: -input-charset not specified, using utf-8 (detected in locale settings)
Total translation table size: 0
Total rockridge attributes bytes: 417
Total directory bytes: 0
Path table size(bytes): 10
Max brk space used 0
184 extents written (0 MB)
zahra@ubuntu:~/bd2c/tp2/nocloud$
```

Figure 18 — Fichier network-config (DHCP sur ens2), instance-id v3 et nouveau seed seed-reseau.iso.

```
zahra@ubuntu:~/bd2c/tp2/nocloud$ sudo virsh destroy bd2c-vm01
sudo virsh undefine bd2c-vm01
sudo cp /var/lib/libvirt/images/ubuntu-template-v1.qcow2 /var/lib/libvirt/images/bd2c-vm01.qcow2

sudo virt-install --connect qemu:///system \
  --name bd2c-vm01 --memory 2048 --vcpus 2 \
  --disk path=/var/lib/libvirt/images/bd2c-vm01.qcow2,format=qcow2,bus=virtio \
  --disk path=/var/lib/libvirt/images/seed-reseau.iso,device=cdrom \
  --network network=default,model=virtio \
  --import --os-variant generic --graphics none --noautoconsole
Domain 'bd2c-vm01' destroyed

Domain 'bd2c-vm01' has been undefined

WARNING  Using --osinfo generic, VM performance may suffer. Specify an accurate OS for optimal results.

Début d'installation…
Création du domaine…
Création de domaine terminée.
zahra@ubuntu:~/bd2c/tp2/nocloud$
```

Figure 19 — Recréation de la VM à partir du template avec le seed seed-reseau.iso.

```
etudiant@bd2c-vm01:~$ ip addr
ip route
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: ens2: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 52:54:00:3e:68:4e brd ff:ff:ff:ff:ff:ff
    altname enp0s2
    inet 192.168.122.138/24 metric 100 brd 192.168.122.255 scope global dynamic ens2
       valid_lft 3511sec preferred_lft 3511sec
    inet6 fe80::5054:ff:fe3e:684e/64 scope link
       valid_lft forever preferred_lft forever
default via 192.168.122.1 dev ens2 proto dhcp src 192.168.122.138 metric 100
192.168.122.0/24 dev ens2 proto kernel scope link src 192.168.122.138 metric 100
192.168.122.1 dev ens2 proto dhcp scope link src 192.168.122.138 metric 100
etudiant@bd2c-vm01:~$
```

Figure 20 — ip addr dans la VM recréée : nouvelle adresse 192.168.122.138 (nouvelle MAC).

```
etudiant@bd2c-vm01:~$ exit
logout
Connection to 192.168.122.138 closed.
zahra@ubuntu:~$ sudo virsh net-dumpxml default
sudo virsh net-dhcp-leases default
<network connections='1'>
  <name>default</name>
  <uuid>8881701d-e5b3-4f06-aec1-9be6a81ed8ad</uuid>
  <forward mode='nat'>
    <nat>
      <port start='1024' end='65535'/>
    </nat>
  </forward>
  <bridge name='virbr0' stp='on' delay='0'/>
  <mac address='52:54:00:41:a9:9d'/>
  <ip address='192.168.122.1' netmask='255.255.255.0'>
    <dhcp>
      <range start='192.168.122.2' end='192.168.122.254'/>
    </dhcp>
  </ip>
</network>

 Expiry Time           MAC address         Protocol   IP address           Hostname    Client ID or DUID
-------------------------------------------------------------------------------------------------------------------------------
-----------------------
 2026-11-06 14:59:31   52:54:00:3e:68:4e   ipv4       192.168.122.138/24   bd2c-vm01   ff:32:39:f9:b5:00:02:00:00:ab:11:
c6:1c:1b:78:28:6f:e0:48

zahra@ubuntu:~$
```

Figure 21 — Définition du réseau « default » (net-dumpxml) et baux DHCP : plage 192.168.122.2 à 192.168.122.254.

```
zahra@ubuntu:~$ ping -c 2 192.168.122.50
PING 192.168.122.50 (192.168.122.50) 56(84) bytes of data.
From 192.168.122.1 icmp_seq=1 Destination Host Unreachable
From 192.168.122.1 icmp_seq=2 Destination Host Unreachable

--- 192.168.122.50 ping statistics ---
2 packets transmitted, 0 received, +2 errors, 100% packet loss, time 1051ms
pipe 2
zahra@ubuntu:~$
```

Figure 22 — Ping vers 192.168.122.50 : hôte injoignable avant la configuration statique.

```
zahra@ubuntu:~$ cd ~/bd2c/tp2/nocloud
cat > network-config <<'EOF'
version: 2
ethernets:
  ens2:
    dhcp4: false
    addresses: [192.168.122.50/24]
    routes:
      - to: default
        via: 192.168.122.1
    nameservers:
      addresses: [192.168.122.1]
EOF
python3 -c "import yaml; yaml.safe_load(open('network-config')); print('YAML OK')"
YAML OK
zahra@ubuntu:~/bd2c/tp2/nocloud$
```

Figure 23 — Fichier network-config en adresse statique (192.168.122.50/24, passerelle et DNS 192.168.122.1) et validation YAML.

```
zahra@ubuntu:~/bd2c/tp2/nocloud$ sed -i 's/^instance-id:.*/instance-id: iid-bd2c-vm01-v4/' meta-data
genisoimage -output ../seed-statique.iso -volid cidata -joliet -rock user-data meta-data network-config
sudo cp ../seed-statique.iso /var/lib/libvirt/images/

sudo virsh destroy bd2c-vm01
sudo virsh undefine bd2c-vm01
sudo cp /var/lib/libvirt/images/ubuntu-template-v1.qcow2 /var/lib/libvirt/images/bd2c-vm01.qcow2

sudo virt-install --connect qemu:///system \
  --name bd2c-vm01 --memory 2048 --vcpus 2 \
  --disk path=/var/lib/libvirt/images/bd2c-vm01.qcow2,format=qcow2,bus=virtio \
  --disk path=/var/lib/libvirt/images/seed-statique.iso,device=cdrom \
  --network network=default,model=virtio \
  --import --os-variant generic --graphics none --noautoconsole
I: -input-charset not specified, using utf-8 (detected in locale settings)
Total translation table size: 0
Total rockridge attributes bytes: 417
Total directory bytes: 0
Path table size(bytes): 10
Max brk space used 0
184 extents written (0 MB)
Domain 'bd2c-vm01' destroyed

Domain 'bd2c-vm01' has been undefined

WARNING  Using --osinfo generic, VM performance may suffer. Specify an accurate OS for optimal results.

Début d'installation…
Création du domaine…                                                    |    0 B  00:00:00
Création de domaine terminée.
zahra@ubuntu:~/bd2c/tp2/nocloud$
```

Figure 24 — Seed seed-statique.iso (instance-id v4) et recréation de la VM.

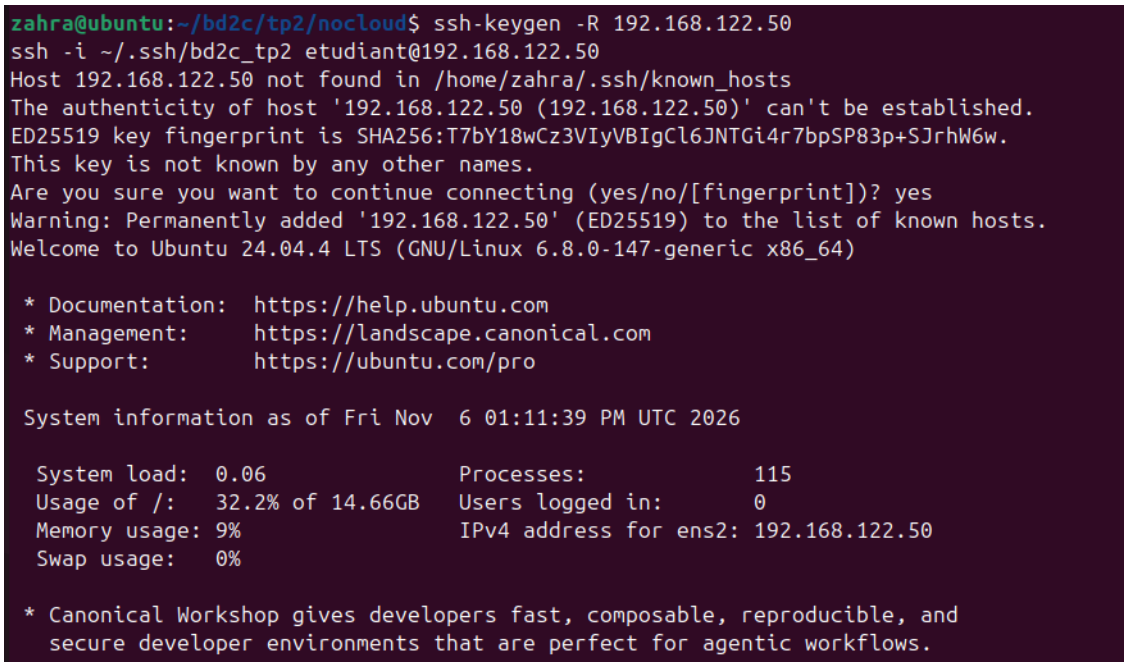


Figure 25 — Suppression de l'ancienne empreinte (ssh-keygen -R) et connexion SSH à 192.168.122.50.

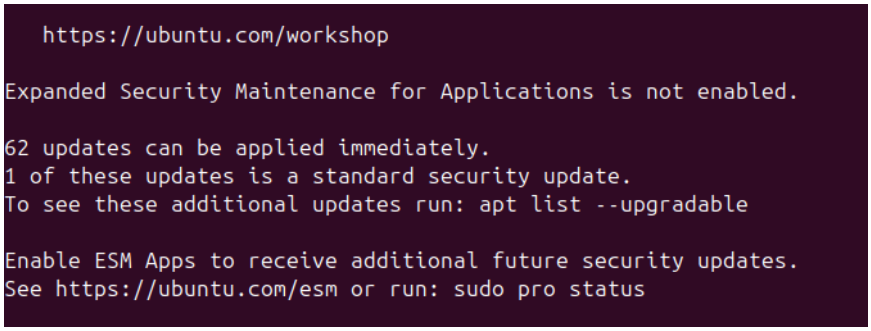


Figure 26 — Fin du message d'accueil de la connexion SSH.

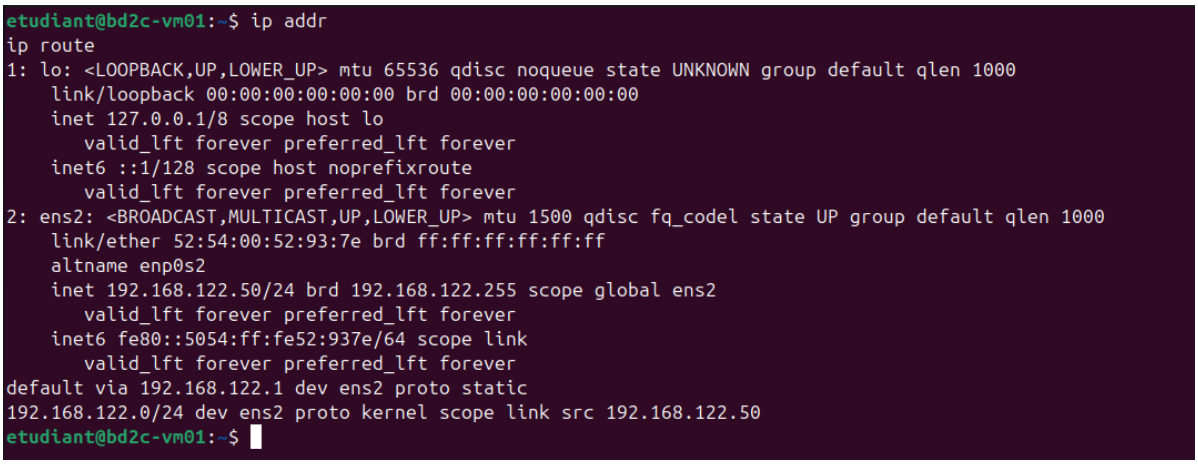


Figure 27 — ip addr : adresse statique 192.168.122.50/24 et route par défaut en « proto static ».

Exercice 7

Installation et configuration de nginx au premier démarrage (paquets, write_files, runcmd).

```
zahra@ubuntu:~$ cd ~/bd2c/tp2/nocloud
cat > user-data <<'EOF'
#cloud-config
hostname: bd2c-vm01
manage_etc_hosts: true
users:
  - default
  - name: etudiant
    shell: /bin/bash
    groups: [sudo]
    sudo: "ALL=(ALL) NOPASSWD:ALL"
    lock_passwd: true
    ssh_authorized_keys:
      - ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIBunbClBfIv91HO2+WiYpAMS3U/2Z96gHA0I5H5lM2am bd2c-tp2
ssh_pwauth: false
package_update: true
packages:
  - nginx
write_files:
  - path: /etc/motd
    content: |
      Laboratoire BD2C – Cloud-Init
    permissions: '0644'
runcmd:
  - [sh, -c, 'systemctl enable --now nginx']
EOF
cat user-data
python3 -c "import yaml; yaml.safe_load(open('user-data')); print('YAML OK')"
#cloud-config
hostname: bd2c-vm01
manage_etc_hosts: true
users:
```

Figure 28 — user-data avec package_update, installation de nginx, écriture de /etc/motd et runcmd.

```
users:
  - default
  - name: etudiant
    shell: /bin/bash
    groups: [sudo]
    sudo: "ALL=(ALL) NOPASSWD:ALL"
    lock_passwd: true
    ssh_authorized_keys:
      - ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIBunbClBfIv91HO2+WiYpAMS3U/2Z96gHA0I5H5lM2am bd2c-tp2
ssh_pwauth: false
package_update: true
packages:
  - nginx
write_files:
  - path: /etc/motd
    content: |
      Laboratoire BD2C – Cloud-Init
    permissions: '0644'
runcmd:
  - [sh, -c, 'systemctl enable --now nginx']
YAML OK
```

Figure 29 — Suite du user-data et validation (« YAML OK »).

```
zahra@ubuntu:~/bd2c/tp2/nocloud$ sed -i 's/^instance-id:.*/instance-id: iid-bd2c-vm01-v5/' meta-data
genisoimage -output ../seed-ex7.iso -volid cidata -joliet -rock user-data meta-data network-config
sudo cp ../seed-ex7.iso /var/lib/libvirt/images/

sudo virsh destroy bd2c-vm01
sudo virsh undefine bd2c-vm01
sudo cp /var/lib/libvirt/images/ubuntu-template-v1.qcow2 /var/lib/libvirt/images/bd2c-vm01.qcow2

sudo virt-install --connect qemu:///system \
  --name bd2c-vm01 --memory 2048 --vcpus 2 \
  --disk path=/var/lib/libvirt/images/bd2c-vm01.qcow2,format=qcow2,bus=virtio \
  --disk path=/var/lib/libvirt/images/seed-ex7.iso,device=cdrom \
  --network network=default,model=virtio \
  --import --os-variant generic --graphics none --noautoconsole
I: -input-charset not specified, using utf-8 (detected in locale settings)
Total translation table size: 0
Total rockridge attributes bytes: 417
Total directory bytes: 0
Path table size(bytes): 10
Max brk space used 0
184 extents written (0 MB)
[sudo] Mot de passe de zahra :
Domain 'bd2c-vm01' destroyed

Domain 'bd2c-vm01' has been undefined

WARNING  Using --osinfo generic, VM performance may suffer. Specify an accurate OS for optimal results.

Début d'installation…
Création du domaine…                                                    |    0 B  00:00:00
Création de domaine terminée.
zahra@ubuntu:~/bd2c/tp2/nocloud$
```

Figure 30 — Seed seed-ex7.iso (instance-id v5) et recréation de la VM.

```
zahra@ubuntu:~/bd2c/tp2/nocloud$ ssh-keygen -R 192.168.122.50
ssh -i ~/.ssh/bd2c_tp2 etudiant@192.168.122.50
# Host 192.168.122.50 found: line 17
# Host 192.168.122.50 found: line 18
# Host 192.168.122.50 found: line 19
/home/zahra/.ssh/known_hosts updated.
Original contents retained as /home/zahra/.ssh/known_hosts.old
The authenticity of host '192.168.122.50 (192.168.122.50)' can't be established.
ED25519 key fingerprint is SHA256:aZVnqauk2jLmI4UtTM9NEZKUAXopkKz4PRTBXRZKZyM.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? YES
Warning: Permanently added '192.168.122.50' (ED25519) to the list of known hosts.
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-147-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Fri Nov  6 07:09:07 PM UTC 2026

  System load:  0.35               Processes:             120
  Usage of /:   32.3% of 14.66GB   Users logged in:       0
  Memory usage: 10%                IPv4 address for ens2: 192.168.122.50
  Swap usage:   0%

 * Canonical Workshop gives developers fast, composable, reproducible, and
   secure developer environments that are perfect for agentic workflows.

   https://ubuntu.com/workshop
```

Figure 31 — Suppression de l'ancienne empreinte et connexion SSH à la nouvelle instance.

```
   https://ubuntu.com/workshop

Expanded Security Maintenance for Applications is not enabled.

80 updates can be applied immediately.
6 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status
```

Figure 32 — Fin du message d'accueil (mises à jour disponibles).

```
etudiant@bd2c-vm01:~$ cloud-init status --wait
status: done
etudiant@bd2c-vm01:~$ systemctl status nginx
curl http://localhost
cat /etc/motd
● nginx.service - A high performance web server and a reverse proxy server
     Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
     Active: active (running) since Fri 2026-11-06 19:07:22 UTC; 4min 56s ago
       Docs: man:nginx(8)
   Main PID: 1444 (nginx)
      Tasks: 3 (limit: 2263)
     Memory: 2.4M (peak: 5.3M)
        CPU: 63ms
     CGroup: /system.slice/nginx.service
             ├─1444 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
             ├─1446 "nginx: worker process"
             └─1447 "nginx: worker process"

Nov 06 19:07:22 bd2c-vm01 systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server>
Nov 06 19:07:22 bd2c-vm01 systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server.
lines 1-15/15 (END)
```

Figure 33 — cloud-init status --wait (done) et systemctl status nginx : service actif (running).

```
etudiant@bd2c-vm01:~$ curl http://localhost
cat /etc/motd
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, the nginx web server is successfully installed and
working. Further configuration is required.</p>

<p>For online documentation and support please refer to
<a href="http://nginx.org/">nginx.org</a>.<br/>
Commercial support is available at
<a href="http://nginx.com/">nginx.com</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
Laboratoire BD2C – Cloud-Init
etudiant@bd2c-vm01:~$
```

Figure 34 — curl http://localhost (page d'accueil nginx) et contenu de /etc/motd (« Laboratoire BD2C — Cloud-Init »).

Exercice 8

Profils applicatifs (web, db, admin).

```
zahra@ubuntu:~/bd2c/tp2/nocloud$ cd ~/bd2c/tp2/profiles
printf 'packages:\n  - nginx\nservice: nginx\nrole: web\n' > web.yml
printf 'packages:\n  - mariadb-server\nservice: mariadb\nrole: db\n' > db.yml
printf 'packages:\n  - htop\nservice: ssh\nrole: admin\n' > admin.yml
cat web.yml db.yml admin.yml
packages:
  - nginx
service: nginx
role: web
packages:
  - mariadb-server
service: mariadb
role: db
packages:
  - htop
service: ssh
role: admin
zahra@ubuntu:~/bd2c/tp2/profiles$
```

Figure 35 — Création des profils web.yml, db.yml et admin.yml (paquets, service et rôle).

Exercice 9

Génération automatique avec Jinja2.

```
zahra@ubuntu:~/bd2c/tp2/profiles$ cd ~/bd2c/tp2
python3 -m venv .venv
source .venv/bin/activate
python -m pip install Jinja2 PyYAML
Collecting Jinja2
  Downloading jinja2-3.1.6-py3-none-any.whl.metadata (2.9 kB)
Collecting PyYAML
  Downloading pyyaml-6.0.3-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl.metadata (2.
4 kB)
Collecting MarkupSafe>=2.0 (from Jinja2)
  Downloading markupsafe-3.0.4-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl.metadata
 (2.7 kB)
Downloading jinja2-3.1.6-py3-none-any.whl (134 kB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 134.9/134.9 kB 2.1 MB/s eta 0:00:00
Downloading pyyaml-6.0.3-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl (807 kB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 807.9/807.9 kB 3.5 MB/s eta 0:00:00
Downloading markupsafe-3.0.4-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl (22 kB)
Installing collected packages: PyYAML, MarkupSafe, Jinja2
Successfully installed Jinja2-3.1.6 MarkupSafe-3.0.4 PyYAML-6.0.3
(.venv) zahra@ubuntu:~/bd2c/tp2$
```

Figure 36 — Environnement virtuel Python (.venv) et installation de Jinja2 et PyYAML.

```
(.venv) zahra@ubuntu:~/bd2c/tp2$ cat > templates/user-data.j2 <<'EOF'
#cloud-config
hostname: {{ hostname }}
manage_etc_hosts: true
users:
  - default
  - name: {{ username }}
    shell: /bin/bash
    groups: [wheel, sudo]
    sudo: "ALL=(ALL) NOPASSWD:ALL"
    lock_passwd: true
    ssh_authorized_keys:
      - {{ ssh_public_key }}
package_update: true
packages:
{% for pkg in packages %}
  - {{ pkg }}
{% endfor %}
EOF
cat templates/user-data.j2
#cloud-config
hostname: {{ hostname }}
manage_etc_hosts: true
users:
  - default
  - name: {{ username }}
    shell: /bin/bash
    groups: [wheel, sudo]
    sudo: "ALL=(ALL) NOPASSWD:ALL"
    lock_passwd: true
    ssh_authorized_keys:
      - {{ ssh_public_key }}
package_update: true
packages:
{% for pkg in packages %}
  - {{ pkg }}
{% endfor %}
(.venv) zahra@ubuntu:~/bd2c/tp2$
```

Figure 37 — Modèle templates/user-data.j2 (hostname, utilisateur, clé publique, liste de paquets).

Exercice 10

Déploiement de trois VM (web, db, admin) à partir d'un inventaire.

```
(.venv) zahra@ubuntu:~/bd2c/tp2$ cat > inventory/vms.yml <<'EOF'
vms:
  - name: bd2c-web
    hostname: web01
    username: etudiant
    profile: web
  - name: bd2c-db
    hostname: db01
    username: etudiant
    profile: db
  - name: bd2c-admin
    hostname: admin01
    username: etudiant
    profile: admin
EOF
(.venv) zahra@ubuntu:~/bd2c/tp2$
```

Figure 38 — Inventaire inventory/vms.yml : bd2c-web, bd2c-db et bd2c-admin.

```
(.venv) zahra@ubuntu:~/bd2c/tp2$ cat > generate.py <<'EOF'
from pathlib import Path
import yaml
from jinja2 import Environment, FileSystemLoader

root = Path(__file__).resolve().parent
inventory = yaml.safe_load((root/'inventory/vms.yml').read_text())
env = Environment(loader=FileSystemLoader(root/'templates'), autoescape=False)
template = env.get_template('user-data.j2')
key = Path.home().joinpath('.ssh/bd2c_tp2.pub').read_text().strip()

for vm in inventory['vms']:
    profile = yaml.safe_load((root/f"profiles/{vm['profile']}.yml").read_text())
    target = root/'generated'/vm['name']
    target.mkdir(parents=True, exist_ok=True)
    (target/'user-data').write_text(template.render(**vm, **profile, ssh_public_key=key))
    (target/'meta-data').write_text(f"instance-id: iid-{vm['name']}\nlocal-hostname: {vm['hostname']}\n")
    print(target)
EOF
(.venv) zahra@ubuntu:~/bd2c/tp2$
```

Figure 39 — Script generate.py : rendu Jinja2 du user-data et du meta-data de chaque VM.

```
(.venv) zahra@ubuntu:~/bd2c/tp2$ python generate.py
find generated -type f | sort
cat generated/bd2c-web/user-data
/home/zahra/bd2c/tp2/generated/bd2c-web
/home/zahra/bd2c/tp2/generated/bd2c-db
/home/zahra/bd2c/tp2/generated/bd2c-admin
generated/bd2c-admin/meta-data
generated/bd2c-admin/user-data
generated/bd2c-db/meta-data
generated/bd2c-db/user-data
generated/bd2c-web/meta-data
generated/bd2c-web/user-data
#cloud-config
hostname: web01
manage_etc_hosts: true
users:
  - default
  - name: etudiant
    shell: /bin/bash
    groups: [wheel, sudo]
    sudo: "ALL=(ALL) NOPASSWD:ALL"
    lock_passwd: true
    ssh_authorized_keys:
      - ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIBunbClBfIv91HO2+WiYpAMS3U/2Z96gHA0I5H5lM2am bd2c-tp2
package_update: true
packages:

  - nginx
(.venv) zahra@ubuntu:~/bd2c/tp2$
```

Figure 40 — Exécution de generate.py : fichiers générés pour chaque VM et user-data de bd2c-web (hostname web01, nginx).

```
(.venv) zahra@ubuntu:~/bd2c/tp2$ for n in web db admin; do
  genisoimage -output generated/seed-bd2c-$n.iso -volid cidata -joliet -rock generated/bd2c-$n/user-data generated/bd2c-$n/meta-data
  sudo cp generated/seed-bd2c-$n.iso /var/lib/libvirt/images/
  sudo cp /var/lib/libvirt/images/ubuntu-template-v1.qcow2 /var/lib/libvirt/images/bd2c-$n.qcow2
done
I: -input-charset not specified, using utf-8 (detected in locale settings)
Total translation table size: 0
Total rockridge attributes bytes: 331
Total directory bytes: 0
Path table size(bytes): 10
Max brk space used 0
183 extents written (0 MB)
I: -input-charset not specified, using utf-8 (detected in locale settings)
Total translation table size: 0
Total rockridge attributes bytes: 331
Total directory bytes: 0
Path table size(bytes): 10
Max brk space used 0
183 extents written (0 MB)
I: -input-charset not specified, using utf-8 (detected in locale settings)
Total translation table size: 0
Total rockridge attributes bytes: 331
Total directory bytes: 0
Path table size(bytes): 10
Max brk space used 0
183 extents written (0 MB)
(.venv) zahra@ubuntu:~/bd2c/tp2$
```

Figure 41 — Génération des trois images seed et copie du template pour chaque VM.

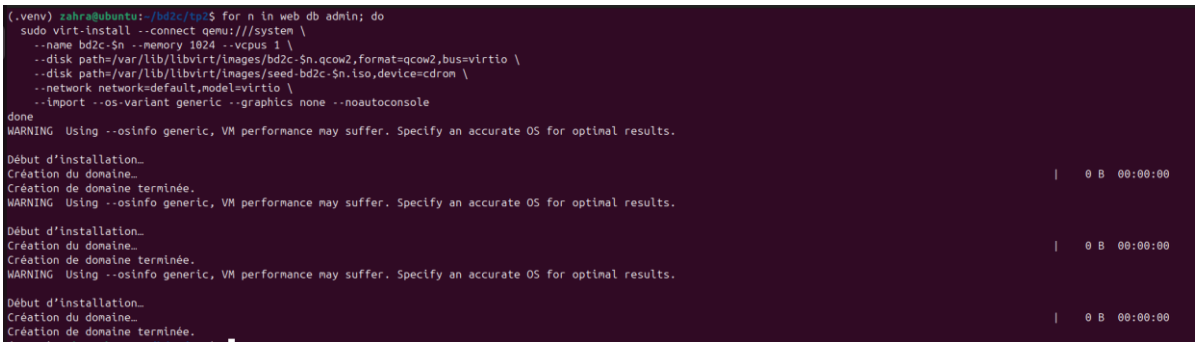


Figure 42 — Création des trois VM avec virt-install.

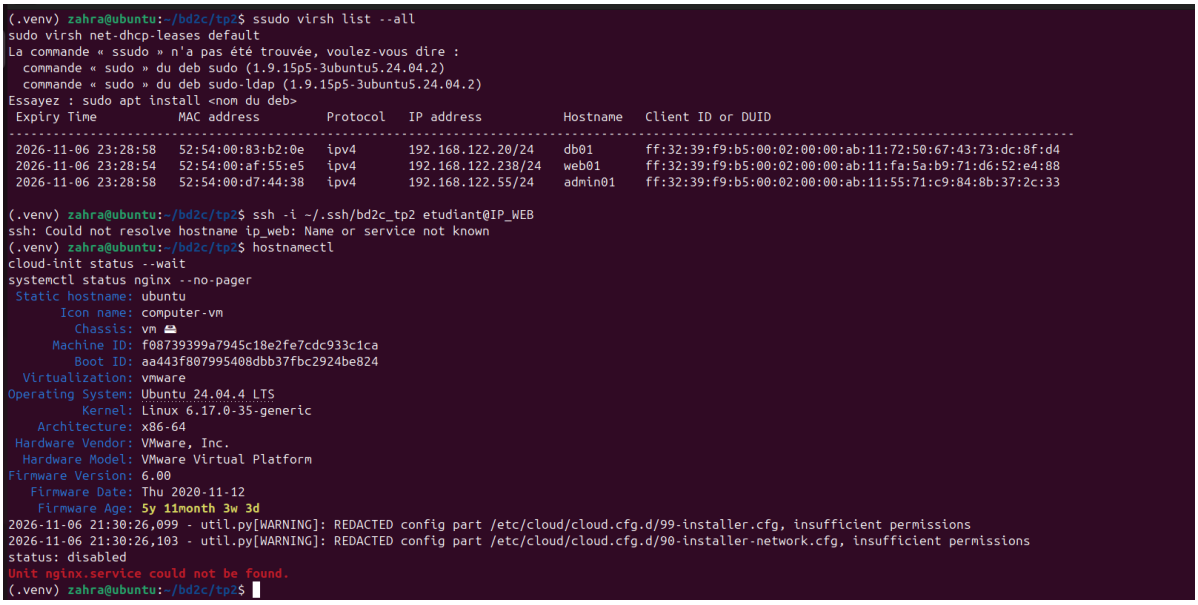


Figure 43 — Baux DHCP des trois VM (db01, web01, admin01). Les commandes de vérification de cette capture ont été lancées sur l'hôte par erreur (nom d'hôte « ubuntu », service nginx introuvable) ; la vérification dans web01 figure sur les captures suivantes.

Exercice 11

11.1 Diagnostic de cloud-init

État, journaux et validation du schéma de cloud-init, puis test d'une erreur de configuration réseau.

```
etudiant@web01:~$ cloud-init status --long
cloud-init analyze show
sudo tail -n 80 /var/log/cloud-init.log
sudo tail -n 80 /var/log/cloud-init-output.log
sudo cloud-init schema --system
status: done
extended_status: done
boot_status_code: enabled-by-generator
last_update: Thu, 01 Jan 1970 00:02:50 +0000
detail: DataSourceNoCloud [seed=/dev/sr0]
errors: []
recoverable_errors: {}
-- Boot Record 01 --
The total time elapsed since completing an event is printed after the "@" character.
The time the event takes is printed after the "+" character.

Starting stage: init-local
|`->no cache found (duration: 0.000s) @00.01300s +00.00100s
|`->found local data from DataSourceNoCloud (duration: 0.096s) @00.02600s +00.09600s
Finished stage: (init-local) 01.25600 seconds

Starting stage: init-network
|`->restored from cache with run check: DataSourceNoCloud [seed=/dev/sr0] (duration: 0.003s) @08.73300s +00.00200s
|`->setting up datasource (duration: 0.000s) @08.80400s +00.00000s
|`->reading and applying user-data (duration: 0.008s) @08.83100s +00.00900s
|`->reading and applying vendor-data (duration: 0.000s) @08.84000s +00.00000s
|`->reading and applying vendor-data2 (duration: 0.000s) @08.84000s +00.00000s
|`->activating datasource (duration: 0.001s) @08.88000s +00.00100s
|`->config-seed_random ran successfully and took 0.000 seconds (duration: 0.001s) @08.91800s +00.00100s
|`->config-growpart ran successfully and took 0.226 seconds (duration: 0.226s) @08.91900s +00.22600s
|`->config-resizefs ran successfully and took 0.052 seconds (duration: 0.052s) @09.14600s +00.05200s
|`->config-mounts ran successfully and took 0.004 seconds (duration: 0.004s) @09.19800s +00.00400s
|`->config-set_hostname ran successfully and took 0.002 seconds (duration: 0.002s) @09.20300s +00.00200s
|`->config-update_hostname ran successfully and took 0.005 seconds (duration: 0.008s) @09.20500s +00.00800s
|`->config-update_etc_hosts ran successfully and took 0.014 seconds (duration: 0.014s) @09.21300s +00.01400s
|`->config-users_groups ran successfully and took 0.168 seconds (duration: 0.169s) @09.22800s +00.16800s
```

Figure 44 — Dans web01 : cloud-init status --long (done, errors: []) et début de cloud-init analyze show.

```
|`->config-update_etc_hosts ran successfully and took 0.014 seconds (duration: 0.014s) @09.21300s +00.01400s
|`->config-users_groups ran successfully and took 0.168 seconds (duration: 0.169s) @09.22800s +00.16800s
|`->config-ssh ran successfully and took 0.805 seconds (duration: 0.805s) @09.39700s +00.80500s
|`->config-set_passwords ran successfully and took 0.001 seconds (duration: 0.001s) @10.20200s +00.00100s
Finished stage: (init-network) 01.49500 seconds

Starting stage: modules-config
|`->config-ssh_import_id ran successfully and took 0.001 seconds (duration: 0.001s) @14.07600s +00.00100s
|`->config-locale ran successfully and took 0.004 seconds (duration: 0.004s) @14.07700s +00.00500s
|`->config-grub_dpkg ran successfully and took 0.567 seconds (duration: 0.568s) @14.08200s +00.56800s
|`->config-apt_configure ran successfully and took 0.031 seconds (duration: 0.031s) @14.65000s +00.03100s
|`->config-byobu ran successfully and took 0.001 seconds (duration: 0.001s) @14.68100s +00.00100s
Finished stage: (modules-config) 00.65400 seconds

Starting stage: modules-final
|`->config-package_update_upgrade_install ran successfully and took 146.078 seconds (duration: 146.079s) @15.43200s +146.51100s
|`->config-reset_rmc ran successfully and took 0.003 seconds (duration: 0.004s) @161.94500s +00.00400s
|`->config-scripts_vendor ran successfully and took 0.002 seconds (duration: 0.002s) @161.94900s +00.00200s
|`->config-scripts_per_once ran successfully and took 0.003 seconds (duration: 0.003s) @161.95100s +00.00400s
|`->config-scripts_per_boot ran successfully and took 0.001 seconds (duration: 0.002s) @161.95500s +00.00200s
|`->config-scripts_per_instance ran successfully and took 0.002 seconds (duration: 0.002s) @161.95700s +00.00200s
|`->config-scripts_user ran successfully and took 0.000 seconds (duration: 0.001s) @161.95900s +00.00100s
|`->config-ssh_authkey_fingerprints ran successfully and took 0.024 seconds (duration: 0.024s) @161.96000s +00.02500s
|`->config-keys_to_console ran successfully and took 0.073 seconds (duration: 0.074s) @161.98500s +00.07400s
|`->config-install_hotplug ran successfully and took 0.003 seconds (duration: 0.003s) @162.05900s +00.00300s
|`->config-final_message ran successfully and took 0.041 seconds (duration: 0.041s) @162.06200s +00.04100s
Finished stage: (modules-final) 146.74900 seconds

Total Time: 150.15400 seconds
```

Figure 45 — Suite de cloud-init analyze show : étapes init-network, modules-config et modules-final (durée totale 150 s).

```
s', '--quiet', 'install', 'nginx'] took 19.638 seconds
2026-11-06 21:31:28,425 - handlers.py[DEBUG]: finish: modules-final/config-package_update_upgrade_install: SUCCESS: config-package_update_upgrade_install ran successfully and took 146.
078 seconds (duration: 146.079s)
2026-11-06 21:31:28,426 - modules.py[DEBUG]: Running module reset_rmc (<module 'cloudinit.config.cc_reset_rmc' from '/usr/lib/python3/dist-packages/cloudinit/config/cc_reset_rmc.py'>)
with frequency once-per-instance
2026-11-06 21:31:28,427 - handlers.py[DEBUG]: start: modules-final/config-reset_rmc: running config-reset_rmc with frequency once-per-instance
2026-11-06 21:31:28,429 - util.py[DEBUG]: Writing to /var/lib/cloud/instances/iid-bd2c-web/sem/config_reset_rmc - wb: [644] 24 bytes
2026-11-06 21:31:28,431 - helpers.py[DEBUG]: Running config-reset_rmc using lock (<FileLock using file '/var/lib/cloud/instances/iid-bd2c-web/sem/config_reset_rmc'>)
2026-11-06 21:31:28,431 - cc_reset_rmc.py[DEBUG]: module disabled, RSCT_PATH not present
2026-11-06 21:31:28,431 - handlers.py[DEBUG]: finish: modules-final/config-reset_rmc: SUCCESS: config-reset_rmc ran successfully and took 0.003 seconds (duration: 0.004s)
2026-11-06 21:31:28,431 - modules.py[DEBUG]: Running module scripts_vendor (<module 'cloudinit.config.cc_scripts_vendor' from '/usr/lib/python3/dist-packages/cloudinit/config/cc_script
s_vendor.py'>) with frequency once-per-instance
2026-11-06 21:31:28,431 - handlers.py[DEBUG]: start: modules-final/config-scripts_vendor: running config-scripts_vendor with frequency once-per-instance
2026-11-06 21:31:28,431 - util.py[DEBUG]: Writing to /var/lib/cloud/instances/iid-bd2c-web/sem/config_scripts_vendor - wb: [644] 24 bytes
2026-11-06 21:31:28,432 - helpers.py[DEBUG]: Running config-scripts_vendor using lock (<FileLock using file '/var/lib/cloud/instances/iid-bd2c-web/sem/config_scripts_vendor'>)
2026-11-06 21:31:28,433 - handlers.py[DEBUG]: finish: modules-final/config-scripts_vendor: SUCCESS: config-scripts_vendor ran successfully and took 0.002 seconds (duration: 0.002s)
2026-11-06 21:31:28,433 - modules.py[DEBUG]: Running module scripts_per_once (<module 'cloudinit.config.cc_scripts_per_once' from '/usr/lib/python3/dist-packages/cloudinit/config/cc_sc
ripts_per_once.py'>) with frequency once
2026-11-06 21:31:28,433 - handlers.py[DEBUG]: start: modules-final/config-scripts_per_once: running config-scripts_per_once with frequency once
2026-11-06 21:31:28,435 - util.py[DEBUG]: Writing to /var/lib/cloud/sem/config_scripts_per_once.once - wb: [644] 23 bytes
2026-11-06 21:31:28,435 - helpers.py[DEBUG]: Running config-scripts_per_once using lock (<FileLock using file '/var/lib/cloud/sem/config_scripts_per_once.once'>)
2026-11-06 21:31:28,437 - handlers.py[DEBUG]: finish: modules-final/config-scripts_per_once: SUCCESS: config-scripts_per_once ran successfully and took 0.003 seconds (duration: 0.003s)
2026-11-06 21:31:28,437 - modules.py[DEBUG]: Running module scripts_per_boot (<module 'cloudinit.config.cc_scripts_per_boot' from '/usr/lib/python3/dist-packages/cloudinit/config/cc_sc
ripts_per_boot.py'>) with frequency always
2026-11-06 21:31:28,437 - handlers.py[DEBUG]: start: modules-final/config-scripts_per_boot: running config-scripts_per_boot with frequency always
2026-11-06 21:31:28,438 - helpers.py[DEBUG]: Running config-scripts_per_boot using lock (<cloudinit.helpers.DummyLock object at 0x7ac9f5c9f1d0>)
2026-11-06 21:31:28,439 - handlers.py[DEBUG]: finish: modules-final/config-scripts_per_boot: SUCCESS: config-scripts_per_boot ran successfully and took 0.001 seconds (duration: 0.002s)
2026-11-06 21:31:28,439 - modules.py[DEBUG]: Running module scripts_per_instance (<module 'cloudinit.config.cc_scripts_per_instance' from '/usr/lib/python3/dist-packages/cloudinit/conf
ig/cc_scripts_per_instance.py'>) with frequency once-per-instance
2026-11-06 21:31:28,439 - handlers.py[DEBUG]: start: modules-final/config-scripts_per_instance: running config-scripts_per_instance with frequency once-per-instance
2026-11-06 21:31:28,439 - util.py[DEBUG]: Writing to /var/lib/cloud/instances/iid-bd2c-web/sem/config_scripts_per_instance - wb: [644] 24 bytes
2026-11-06 21:31:28,440 - helpers.py[DEBUG]: Running config-scripts_per_instance using lock (<FileLock using file '/var/lib/cloud/instances/iid-bd2c-web/sem/config_scripts_per_instance
'>)
2026-11-06 21:31:28,441 - handlers.py[DEBUG]: finish: modules-final/config-scripts_per_instance: SUCCESS: config-scripts_per_instance ran successfully and took 0.002 seconds (duration:
 0.002s)
2026-11-06 21:31:28,441 - modules.py[DEBUG]: Running module scripts_user (<module 'cloudinit.config.cc_scripts_user' from '/usr/lib/python3/dist-packages/cloudinit/config/cc_scripts_us
er.py'>) with frequency once-per-instance
2026-11-06 21:31:28,441 - handlers.py[DEBUG]: start: modules-final/config-scripts_user: running config-scripts_user with frequency once-per-instance
```

Figure 46 — Extrait de /var/log/cloud-init.log (modules final).

```
nit/config/cc_ssh_authkey_fingerprints.py'>) with frequency once-per-instance
2026-11-06 21:31:28,442 - handlers.py[DEBUG]: start: modules-final/config-ssh_authkey_fingerprints: running config-ssh_authkey_fingerprints with frequency once-per-instance
2026-11-06 21:31:28,442 - util.py[DEBUG]: Writing to /var/lib/cloud/instances/iid-bd2c-web/sem/config_ssh_authkey_fingerprints - wb: [644] 24 bytes
2026-11-06 21:31:28,443 - helpers.py[DEBUG]: Running config-ssh_authkey_fingerprints using lock (<FileLock using file '/var/lib/cloud/instances/iid-bd2c-web/sem/config_ssh_authkey_fing
erprints'>)
2026-11-06 21:31:28,445 - util.py[DEBUG]: Reading from /etc/ssh/sshd_config (quiet=False)
2026-11-06 21:31:28,445 - util.py[DEBUG]: Reading 3517 bytes from /etc/ssh/sshd_config
2026-11-06 21:31:28,446 - util.py[DEBUG]: Reading from /home/etudiant/.ssh/authorized_keys (quiet=False)
2026-11-06 21:31:28,448 - util.py[DEBUG]: Reading 90 bytes from /home/etudiant/.ssh/authorized_keys
2026-11-06 21:31:28,464 - util.py[DEBUG]: Reading from /etc/ssh/sshd_config (quiet=False)
2026-11-06 21:31:28,464 - util.py[DEBUG]: Reading 3517 bytes from /etc/ssh/sshd_config
2026-11-06 21:31:28,464 - util.py[DEBUG]: Reading from /home/ubuntu/.ssh/authorized_keys (quiet=False)
2026-11-06 21:31:28,465 - util.py[DEBUG]: Reading 0 bytes from /home/ubuntu/.ssh/authorized_keys
2026-11-06 21:31:28,467 - handlers.py[DEBUG]: finish: modules-final/config-ssh_authkey_fingerprints: SUCCESS: config-ssh_authkey_fingerprints ran successfully and took 0.024 seconds (d
uration: 0.024s)
2026-11-06 21:31:28,467 - modules.py[DEBUG]: Running module keys_to_console (<module 'cloudinit.config.cc_keys_to_console' from '/usr/lib/python3/dist-packages/cloudinit/config/cc_keys
_to_console.py'>) with frequency once-per-instance
2026-11-06 21:31:28,467 - handlers.py[DEBUG]: start: modules-final/config-keys_to_console: running config-keys_to_console with frequency once-per-instance
2026-11-06 21:31:28,467 - util.py[DEBUG]: Writing to /var/lib/cloud/instances/iid-bd2c-web/sem/config_keys_to_console - wb: [644] 24 bytes
2026-11-06 21:31:28,468 - helpers.py[DEBUG]: Running config-keys_to_console using lock (<FileLock using file '/var/lib/cloud/instances/iid-bd2c-web/sem/config_keys_to_console'>)
2026-11-06 21:31:28,468 - subp.py[DEBUG]: Running command ['/usr/lib/cloud-init/write-ssh-key-fingerprints', '', ''] with allowed return codes [0] (shell=False, capture=True)
2026-11-06 21:31:28,515 - performance.py[DEBUG]: Running ['/usr/lib/cloud-init/write-ssh-key-fingerprints', '', ''] took 0.047 seconds
2026-11-06 21:31:28,541 - performance.py[DEBUG]: Writing to console took 0.025 seconds
2026-11-06 21:31:28,541 - handlers.py[DEBUG]: finish: modules-final/config-keys_to_console: SUCCESS: config-keys_to_console ran successfully and took 0.073 seconds (duration: 0.074s)
2026-11-06 21:31:28,541 - modules.py[DEBUG]: Running module install_hotplug (<module 'cloudinit.config.cc_install_hotplug' from '/usr/lib/python3/dist-packages/cloudinit/config/cc_inst
all_hotplug.py'>) with frequency once-per-instance
2026-11-06 21:31:28,541 - handlers.py[DEBUG]: start: modules-final/config-install_hotplug: running config-install_hotplug with frequency once-per-instance
2026-11-06 21:31:28,541 - util.py[DEBUG]: Writing to /var/lib/cloud/instances/iid-bd2c-web/sem/config_install_hotplug - wb: [644] 24 bytes
2026-11-06 21:31:28,542 - helpers.py[DEBUG]: Running config-install_hotplug using lock (<FileLock using file '/var/lib/cloud/instances/iid-bd2c-web/sem/config_install_hotplug'>)
2026-11-06 21:31:28,542 - util.py[DEBUG]: Reading from /var/lib/cloud/hotplug.enabled (quiet=False)
2026-11-06 21:31:28,542 - util.py[DEBUG]: File not found: /var/lib/cloud/hotplug.enabled
2026-11-06 21:31:28,544 - stages.py[DEBUG]: Allowed events: {<EventScope.NETWORK: 'network'>: {<EventType.BOOT_NEW_INSTANCE: 'boot-new-instance'>}}
2026-11-06 21:31:28,544 - stages.py[DEBUG]: Event Denied: scopes=['network'] EventType=hotplug
2026-11-06 21:31:28,544 - cc_install_hotplug.py[DEBUG]: Skipping hotplug install, not enabled
2026-11-06 21:31:28,544 - handlers.py[DEBUG]: finish: modules-final/config-install_hotplug: SUCCESS: config-install_hotplug ran successfully and took 0.003 seconds (duration: 0.003s)
2026-11-06 21:31:28,544 - modules.py[DEBUG]: Running module final_message (<module 'cloudinit.config.cc_final_message' from '/usr/lib/python3/dist-packages/cloudinit/config/cc_final_me
```

Figure 47 — Suite du journal /var/log/cloud-init.log.

```
2026-11-06 21:31:28,544 - handlers.py[DEBUG]: finish: modules-final/config-install_hotplug: SUCCESS: config-install_hotplug ran successfully and took 0.003 seconds (duration: 0.003s)
2026-11-06 21:31:28,544 - modules.py[DEBUG]: Running module final_message (<module 'cloudinit.config.cc_final_message' from '/usr/lib/python3/dist-packages/cloudinit/config/cc_final_me
ssage.py'>) with frequency always
2026-11-06 21:31:28,544 - handlers.py[DEBUG]: start: modules-final/config-final_message: running config-final_message with frequency always
2026-11-06 21:31:28,544 - helpers.py[DEBUG]: Running config-final_message using lock (<cloudinit.helpers.DummyLock object at 0x7ac9f5c9f6e0>)
2026-11-06 21:31:28,545 - util.py[DEBUG]: Reading from /proc/uptime (quiet=False)
2026-11-06 21:31:28,545 - util.py[DEBUG]: Reading 14 bytes from /proc/uptime
2026-11-06 21:31:28,580 - performance.py[DEBUG]: Rendering jinja2 template took 0.035 seconds
2026-11-06 21:31:28,584 - log_util.py[DEBUG]: Cloud-init v. 26.2-0ubuntu1~24.04.2 finished at Fri, 06 Nov 2026 21:31:28 +0000. Datasource DataSourceNoCloud [seed=/dev/sr0].  Up 170.91
seconds
2026-11-06 21:31:28,584 - util.py[DEBUG]: Writing to /var/lib/cloud/instance/boot-finished - wb: [644] 68 bytes
2026-11-06 21:31:28,585 - handlers.py[DEBUG]: finish: modules-final/config-final_message: SUCCESS: config-final_message ran successfully and took 0.041 seconds (duration: 0.041s)
2026-11-06 21:31:28,586 - main.py[DEBUG]: Ran 11 modules with 0 failures
2026-11-06 21:31:28,586 - util.py[DEBUG]: Reading from /proc/uptime (quiet=False)
2026-11-06 21:31:28,587 - util.py[DEBUG]: Reading 14 bytes from /proc/uptime
2026-11-06 21:31:28,590 - atomic_helper.py[DEBUG]: Atomically writing to file /var/lib/cloud/data/status.json (via temporary file /var/lib/cloud/data/tmpfhq6tj2x) - w: [644] 522 bytes/
chars
2026-11-06 21:31:28,591 - atomic_helper.py[DEBUG]: Atomically writing to file /var/lib/cloud/data/result.json (via temporary file /var/lib/cloud/data/tmpnhzkzh1j) - w: [644] 84 bytes/c
hars
2026-11-06 21:31:28,591 - util.py[DEBUG]: Creating symbolic link from '/run/cloud-init/result.json' => '../../var/lib/cloud/data/result.json'
2026-11-06 21:31:28,592 - performance.py[DEBUG]: cloud-init stage: 'modules-final' took 146.426 seconds
2026-11-06 21:31:28,592 - handlers.py[DEBUG]: finish: modules-final: SUCCESS: running modules for final (duration: 146.426s)
|          .o==*=oo|
|           +=o*=.o|
|        E .=+...  |
|        S+ .+..   |
|       .. .o..    |
|        + oo. o   |
|       o +.. . = |
|      ..o  ...o o|
+----[SHA256]-----+
Cloud-init v. 26.2-0ubuntu1~24.04.2 running 'modules:config' at Fri, 06 Nov 2026 21:29:00 +0000. Up 23.17 seconds.
Cloud-init v. 26.2-0ubuntu1~24.04.2 running 'modules:final' at Fri, 06 Nov 2026 21:29:01 +0000. Up 24.57 seconds.
Hit:1 http://ma.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:3 http://ma.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:4 http://ma.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:5 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1,157 kB]
```

Figure 48 — Fin de cloud-init.log et début de cloud-init-output.log (mise à jour des paquets apt).

```
Get:5 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1,157 kB]
Get:6 http://ma.archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1,425 kB]
Get:7 http://ma.archive.ubuntu.com/ubuntu noble-updates/main Translation-en [313 kB]
Get:8 http://ma.archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [181 kB]
Get:9 http://ma.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [1,840 kB]
Get:10 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [234 kB]
Get:11 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [46.5 kB]
Get:12 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [1,721 kB]
Get:13 http://security.ubuntu.com/ubuntu noble-security/restricted Translation-en [394 kB]
Get:14 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [1,228 kB]
Get:15 http://security.ubuntu.com/ubuntu noble-security/universe Translation-en [248 kB]
Get:16 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [76.2 kB]
Get:17 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Packages [45.4 kB]
Get:18 http://security.ubuntu.com/ubuntu noble-security/multiverse Translation-en [12.4 kB]
Get:19 http://ma.archive.ubuntu.com/ubuntu noble-updates/restricted Translation-en [420 kB]
Get:20 http://ma.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1,708 kB]
Get:21 http://ma.archive.ubuntu.com/ubuntu noble-updates/universe Translation-en [343 kB]
Get:22 http://ma.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [388 kB]
Get:23 http://ma.archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Packages [50.7 kB]
Get:24 http://ma.archive.ubuntu.com/ubuntu noble-updates/multiverse Translation-en [14.3 kB]
Get:25 http://ma.archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
Get:26 http://ma.archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [5,760 B]
Get:27 http://ma.archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [12.6 kB]
Fetched 12.2 MB in 1min 48s (113 kB/s)
Reading package lists...
Reading package lists...
Building dependency tree...
Reading state information...
The following additional packages will be installed:
  nginx-common
Suggested packages:
  fcgiwrap nginx-doc ssl-cert
The following NEW packages will be installed:
  nginx nginx-common
0 upgraded, 2 newly installed, 0 to remove and 72 not upgraded.
Need to get 570 kB of archives.
After this operation, 1,602 kB of additional disk space will be used.
```

Figure 49 — cloud-init-output.log : installation des paquets nginx et nginx-common.

```
Get:1 http://ma.archive.ubuntu.com/ubuntu noble-updates/main amd64 nginx-common all 1.24.0-2ubuntu7.19 [45.3 kB]
Get:2 http://ma.archive.ubuntu.com/ubuntu noble-updates/main amd64 nginx amd64 1.24.0-2ubuntu7.19 [525 kB]
Preconfiguring packages ...
Fetched 570 kB in 6s (100 kB/s)
Selecting previously unselected package nginx-common.
(Reading database ... 88198 files and directories currently installed.)
Preparing to unpack .../nginx-common_1.24.0-2ubuntu7.19_all.deb ...
Unpacking nginx-common (1.24.0-2ubuntu7.19) ...
Selecting previously unselected package nginx.
Preparing to unpack .../nginx_1.24.0-2ubuntu7.19_amd64.deb ...
Unpacking nginx (1.24.0-2ubuntu7.19) ...
Setting up nginx-common (1.24.0-2ubuntu7.19) ...
Created symlink /etc/systemd/system/multi-user.target.wants/nginx.service → /usr/lib/systemd/system/nginx.service.
Setting up nginx (1.24.0-2ubuntu7.19) ...
 * Upgrading binary nginx                                     [ OK ]
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for ufw (0.36.2-6) ...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
Cloud-init v. 26.2-0ubuntu1~24.04.2 finished at Fri, 06 Nov 2026 21:31:28 +0000. Datasource DataSourceNoCloud [seed=/dev/sr0].  Up 170.91 seconds
Found cloud-config data types: user-data, network-config

1. user-data at /var/lib/cloud/instances/iid-bd2c-web/cloud-config.txt:
  Valid schema user-data

2. network-config at /var/lib/cloud/instances/iid-bd2c-web/network-config.json:
  Valid schema network-config
etudiant@web01:~$
```

Figure 50 — Fin du journal de sortie et cloud-init schema --system : user-data et network-config valides.

```
(.venv) zahra@ubuntu:~/bd2c/tp2$ cd ~/bd2c/tp2/nocloud
cat > network-config <<'EOF'
version: 2
ethernets:
  enp9s9:
    dhcp4: true
EOF
sed -i 's/^instance-id:.*/instance-id: iid-bd2c-vm01-erreur/' meta-data
genisoimage -output ../seed-erreur.iso -volid cidata -joliet -rock user-data meta-data network-config
sudo cp ../seed-erreur.iso /var/lib/libvirt/images/
I: -input-charset not specified, using utf-8 (detected in locale settings)
Total translation table size: 0
Total rockridge attributes bytes: 417
Total directory bytes: 0
Path table size(bytes): 10
Max brk space used 0
184 extents written (0 MB)
[sudo] Mot de passe de zahra :
(.venv) zahra@ubuntu:~/bd2c/tp2/nocloud$
```

Figure 51 — Test d'erreur : network-config avec une interface inexistante (enp9s9) et seed seed-erreur.iso.

```
(.venv) zahra@ubuntu:~/bd2c/tp2/nocloud$ sudo virsh destroy bd2c-vm01
sudo virsh undefine bd2c-vm01
sudo cp /var/lib/libvirt/images/ubuntu-template-v1.qcow2 /var/lib/libvirt/images/bd2c-vm01.qcow2

sudo virt-install --connect qemu:///system \
  --name bd2c-vm01 --memory 1024 --vcpus 1 \
  --disk path=/var/lib/libvirt/images/bd2c-vm01.qcow2,format=qcow2,bus=virtio \
  --disk path=/var/lib/libvirt/images/seed-erreur.iso,device=cdrom \
  --network network=default,model=virtio \
  --import --os-variant generic --graphics none --noautoconsole
error: Failed to destroy domain 'bd2c-vm01'
error: Requested operation is not valid: domain is not running

Domain 'bd2c-vm01' has been undefined

WARNING  Using --osinfo generic, VM performance may suffer. Specify an accurate OS for optimal results.

Début d'installation…
Création du domaine…                                                    |    0 B  00:00:00
Création de domaine terminée.
(.venv) zahra@ubuntu:~/bd2c/tp2/nocloud$
```

Figure 52 — Recréation de bd2c-vm01 avec le seed erroné (l'arrêt a échoué car la VM n'était pas en cours d'exécution).

11.2 Formats d'images RAW et QCOW2

Création, inspection, conversion et comparaison de disques vides de 5 Gio.

```
zahra@ubuntu:~$ cd ~/tp1-kvm/formats
du -h disque-raw.img
qemu-img convert -p -f raw -O qcow2 disque-raw.img disque-converti.qcow2
qemu-img info disque-converti.qcow2
qemu-img compare -f raw -F qcow2 disque-raw.img disque-converti.qcow2
4,0K    disque-raw.img
    (100.00/100%)
image: disque-converti.qcow2
file format: qcow2
virtual size: 5 GiB (5368709120 bytes)
disk size: 196 KiB
cluster_size: 65536
Format specific information:
    compat: 1.1
    compression type: zlib
    lazy refcounts: false
    refcount bits: 16
    corrupt: false
    extended l2: false
Child node '/file':
    filename: disque-converti.qcow2
    protocol type: file
    file length: 192 KiB (197120 bytes)
    disk size: 196 KiB
Images are identical.
zahra@ubuntu:~/tp1-kvm/formats$
```

Figure 53 — Conversion du disque RAW en QCOW2 (qemu-img convert), informations sur le disque converti et comparaison du contenu logique : « Images are identical. »

Comparaison des formats (valeurs observées)

Fichier	Format	Taille virtuelle	Taille apparente (ls -lh)	Occupation réelle
disque-raw.img	raw	5 GiB	5,0 G	4 KiB
disque-qcow2.qcow2	qcow2	5 GiB	193 K	196 KiB
disque-converti.qcow2	qcow2	5 GiB	192 KiB	196 KiB

Questions

1. **Pourquoi un RAW de 5 Gio peut-il occuper bien moins de 5 Gio sur l'hôte ?** C'est un fichier sparse : les zones jamais écrites ne sont pas allouées. Ici, ls annonce 5,0 G alors que du affiche 4 KiB.
2. **Pourquoi « RAW occupe toujours toute sa capacité » est-il faux ?** Cela dépend de la création et du système de fichiers : le RAW vide de ce TP n'occupe que 4 KiB. Il occuperait toute sa capacité s'il était préalloué.
3. **Pourquoi le format QCOW2 ne signifie-t-il pas qu'un OS est installé ?** C'est seulement un format de conteneur. disque-qcow2.qcow2 est au format QCOW2 mais vide, alors que la capture de l'exercice 7 est aussi en QCOW2 et contient Ubuntu.
4. **Convertir une image la généralise-t-il ?** Non : convert copie le contenu tel quel (machine-id, clés SSH et hostname inclus). La généralisation se fait avec cloud-init et virt-sysprep.
5. **Un template peut-il être stocké en RAW ? Justifier.** Oui, techniquement, mais le QCOW2 est plus adapté : il est plus compact (allocation progressive) et prend en charge les backing files et les snapshots. Le RAW est plus simple mais plus lourd à stocker et à transférer.
6. **Pourquoi la conversion d'un disque système doit-elle être réalisée hors ligne ?** Si la VM tourne, elle écrit sur le disque pendant la copie : l'image obtenue peut être incohérente ou corrompue. La VM est donc arrêtée d'abord (comme vm01 dans ce TP).

Exercice 12 — Études de cas

Cas 1 — Tester une mise à jour risquée sur une copie d'un serveur existant

- **Choix principal :** clone complet, dans un environnement isolé.
- **Justification :** le clone a un disque indépendant : la mise à jour ne touche pas le serveur d'origine. Un snapshot peut compléter un test local mais n'apporte pas la même indépendance.
- **Risque :** le clone garde les mêmes identités et la même adresse IP que l'original ; il peut créer des conflits sur le réseau ou contacter les vrais services.
- **Vérification :** qemu-img info (pas de backing file), réseau isolé avant le démarrage, puis comparaison du hostname, du machine-id et de l'adresse IP.

Cas 2 — Fournir vingt VM Linux identiques en logiciels, mais avec des noms et identités distincts

- **Choix principal :** template généralisé avec personnalisation par instance (cloud-init).
- **Justification :** même base logicielle ; chaque VM reçoit son hostname, son machine-id, ses clés SSH et son instance-id au premier démarrage.
- **Risque :** un template mal nettoyé reproduit les mêmes identités sur les vingt VM.

- **Vérification :** comparer hostname, machine-id, clé SSH hôte et instance-id de chaque instance, comme dans le tableau app01/app02 de l'exercice 10.

Cas 3 — Conserver le système installé d'une VM avant une migration

- **Choix principal :** image d'OS cohérente (capture du disque), avec la configuration de la VM et ses dépendances.
- **Justification :** elle conserve le système installé et son contenu tel quel, sans le généraliser.
- **Risque :** une image seule ne prouve pas que le système démarre ni que les données sont cohérentes ; le XML seul ne suffit pas à reconstruire la VM.
- **Vérification :** test de réimportation obligatoire (comme à l'exercice 8), avec somme de contrôle et qemu-img check.

Cas 4 — Un fichier QCOW2 vide, et un serveur Ubuntu opérationnel à obtenir

- **Choix principal :** installer depuis une ISO, ou partir d'une image d'OS compatible.
- **Justification :** un disque vide n'est pas amorçable : il ne contient ni système ni chargeur de démarrage.
- **Risque :** croire que le format QCOW2 suffit : le disque-qcow2.qcow2 de l'exercice 11 est valide mais vide.
- **Vérification :** la VM démarre et /etc/os-release s'affiche.

Cas 5 — Transmettre une base Linux préinstallée et réutilisable à plusieurs équipes

- **Choix principal :** template ou image généralisée, documentée, auditée et versionnée.
- **Justification :** chaque équipe déploie ses instances à partir de la même référence, accompagnée d'une fiche (OS, logiciels, nettoyage, somme de contrôle).
- **Risque :** diffuser un compte, un mot de passe, une clé ou des données personnelles restés dans l'image ; le nettoyage ne supprime pas tout automatiquement.
- **Vérification :** audit du contenu (comptes, clés, historique), puis somme SHA-256 et test de déploiement.

Cas 6 — Restaurer une application utilisant trois disques et une base de données

- **Choix principal :** sauvegarde cohérente de l'ensemble (les trois disques, la base de données et la configuration), et non une simple copie du disque système.
- **Justification :** les données applicatives et la configuration sont nécessaires pour restaurer l'application.
- **Risque :** copier les disques à des moments différents, ou pendant que la base écrit, donne un état incohérent.
- **Vérification :** test de restauration complet, puis contrôle de l'application et de la base.

Cas 7 — Créer rapidement cinq VM temporaires en minimisant le stockage local

- **Choix principal :** éventuellement des clones liés (backing file) sur une référence immuable.
- **Justification :** chaque overlay ne stocke que ses différences, donc le stockage reste faible.

- **Risque :** la dépendance à la chaîne de backing files : si la base est modifiée ou supprimée, tous les overlays deviennent inutilisables, et déplacer seulement l'overlay ne déplace pas la VM.
- **Vérification :** qemu-img info --backing-chain, base en lecture seule, et aplatissement avec qemu-img convert si une image autonome est nécessaire.

Cas 8 — VM clonée avec une nouvelle MAC mais le même hostname et la même clé SSH hôte : le clonage a-t-il échoué ?

- **Choix principal :** non, ce n'est pas un échec.
- **Justification :** virt-clone renouvelle les identités du matériel virtuel (UUID, MAC) mais pas celles de l'OS (hostname, machine-id, clés SSH), qui sont dans le disque, comme observé à l'exercice 6.
- **Risque :** deux serveurs indiscernables et des conflits d'adresse IP ou de DHCP (à l'exercice 6, les deux VM ont obtenu 192.168.122.66).
- **Vérification :** comparer UUID, MAC, hostname, machine-id et clé SSH ; pour corriger, générer les identités par instance (cloud-init, virt-sysprep).